Project Report

on

Smart Route Optimization System

Submitted by

Parshant Katoch
25MCA20178

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Deepali Saini

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1.Objective

The main objective of this project is to develop a Flight Route Optimization System that helps users find the most efficient flight path between two locations using graph-based algorithms.

The system aims to provide an interactive and user-friendly interface to visualize routes on a map, analyze different flight options based on distance, cost, and time, and improve decision-making for travel planning.

Additionally, the project focuses on implementing advanced algorithms like Dijkstra and A* to compute optimal paths and display real-time route analytics.

2.Introduction

In modern transportation systems, finding the optimal flight route is a complex task due to multiple factors such as distance, cost, and travel time. Traditional systems often fail to provide efficient and dynamic route optimization.

This project presents a Flight Route Optimization System developed using modern web technologies. The system is designed to visualize flight networks as graphs, where airports are nodes and routes are edges.

By applying graph algorithms like Dijkstra and A*, the system calculates the most efficient routes. The application also provides interactive map visualization, real-time analytics, and route comparison features, making it a powerful tool for both users and administrators.

The project follows a modular architecture and aims to simulate real-world flight route planning systems.

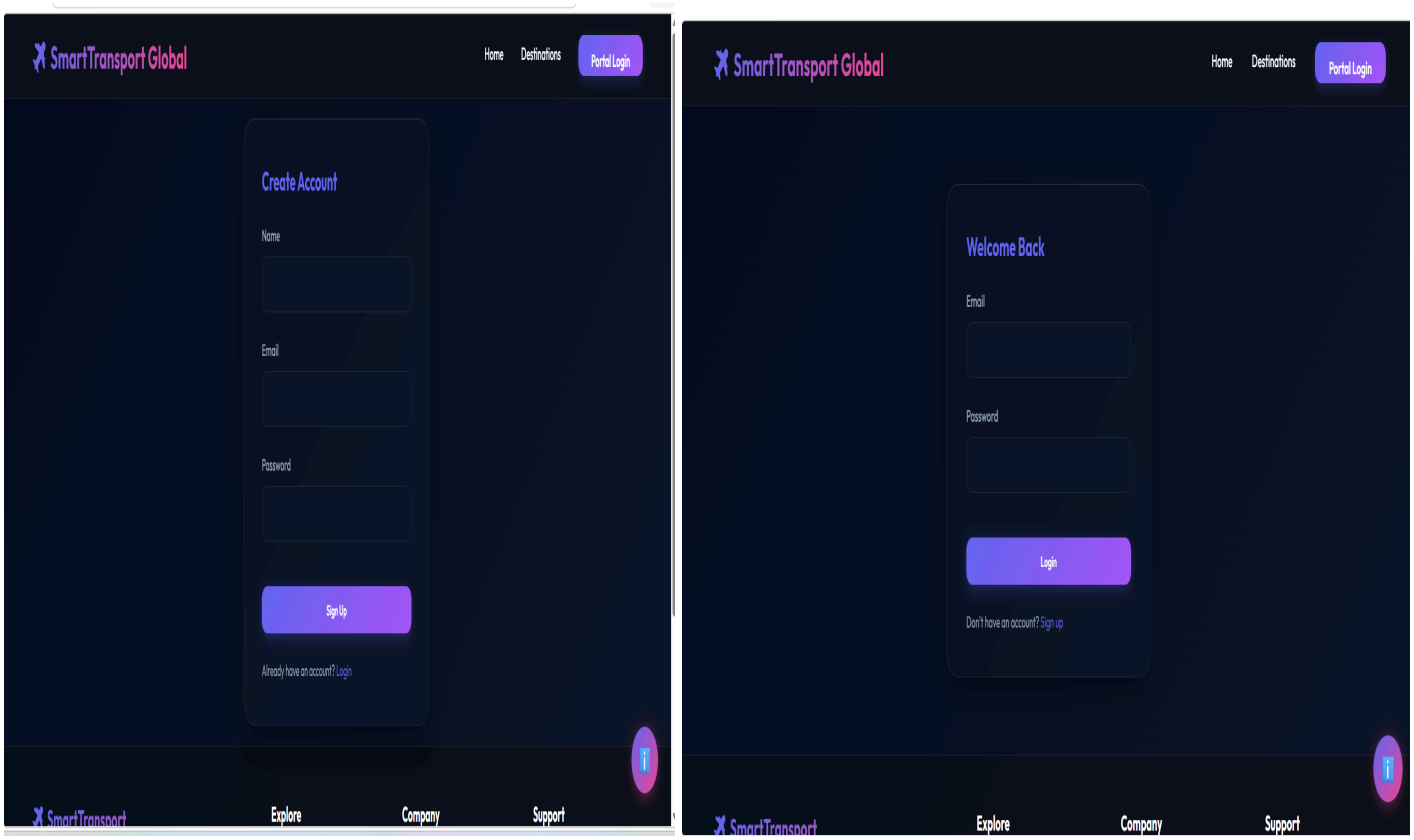
3.Tools and Technologies Used

Category	Technology	Purpose / Description
Frontend Technologies	React.js	For building interactive user interface.
	React-Leaflet	For map visualization.
	Chart.js	For analytics dashboard
	React Router	For Navigation
Backend Technologies	Node.js & Express.js	For building RESTful APIs
	MongoDB & Mongoose	For database management
Algorithms	Dijkstra Algorithm	For shortest path calculation

	A* Algorithm	For optimized route finding
Other Tools	JWT Bcrypt	For authentication For password security

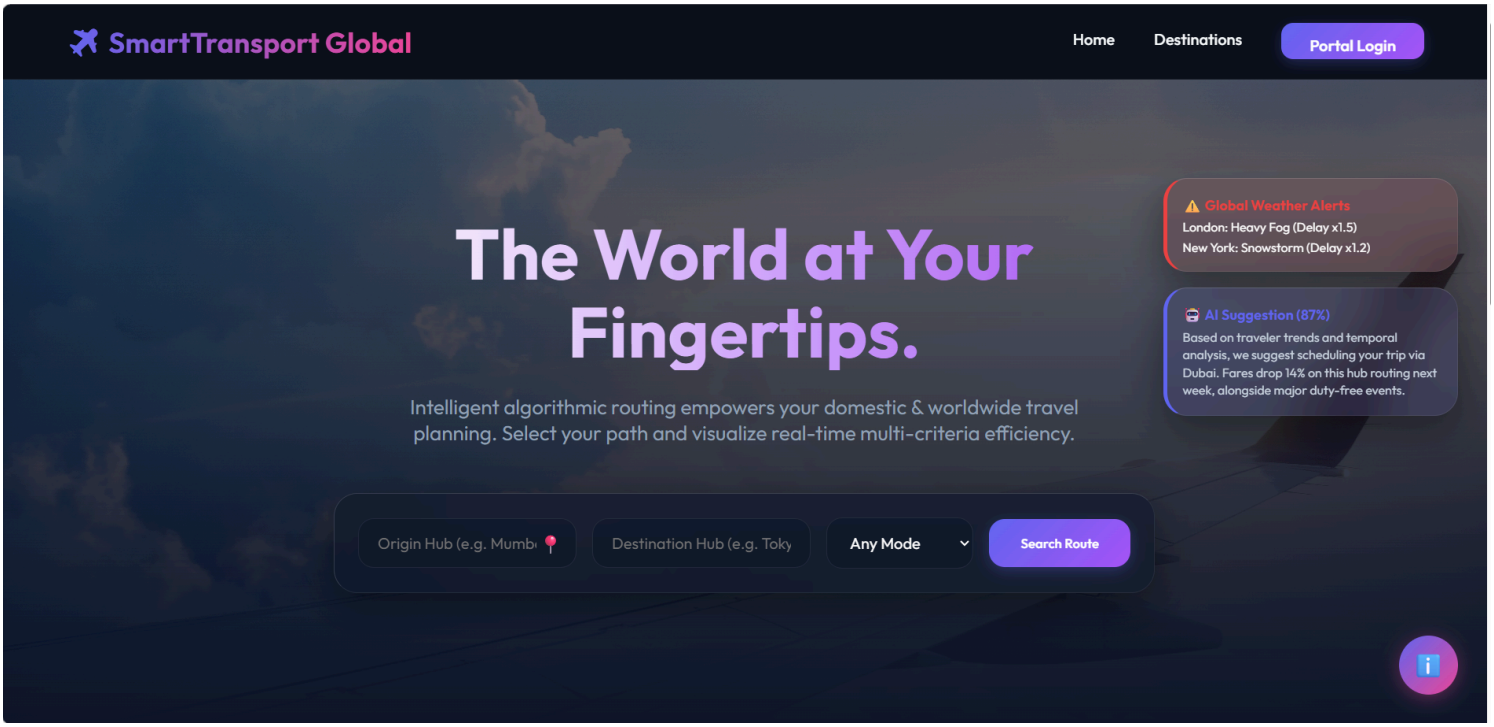
4. Input Screenshots

Figure 1



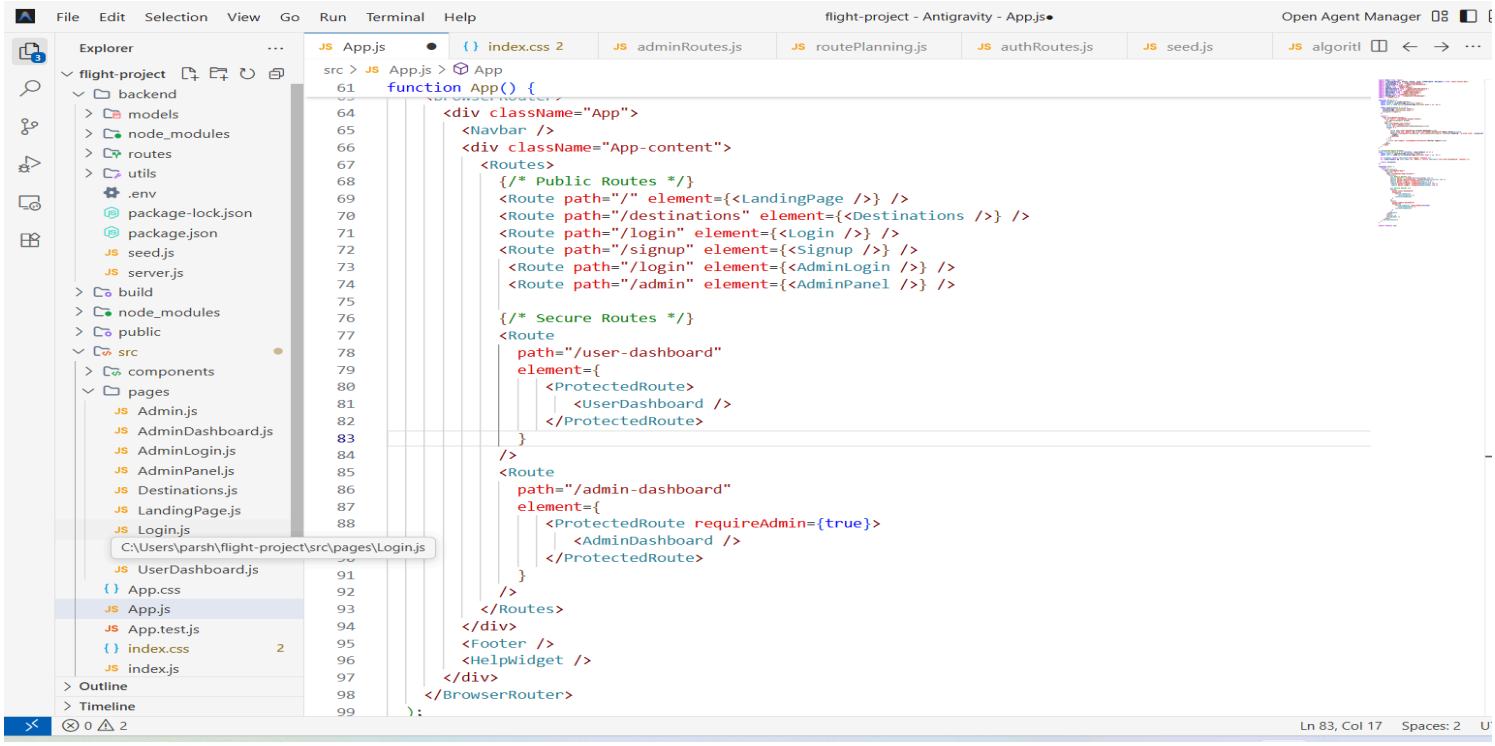
User Authentication Interface. This form captures the user's initial registration details (Name, Email, Password, Company) securely hashing the data before saving it to the database

Figure 2



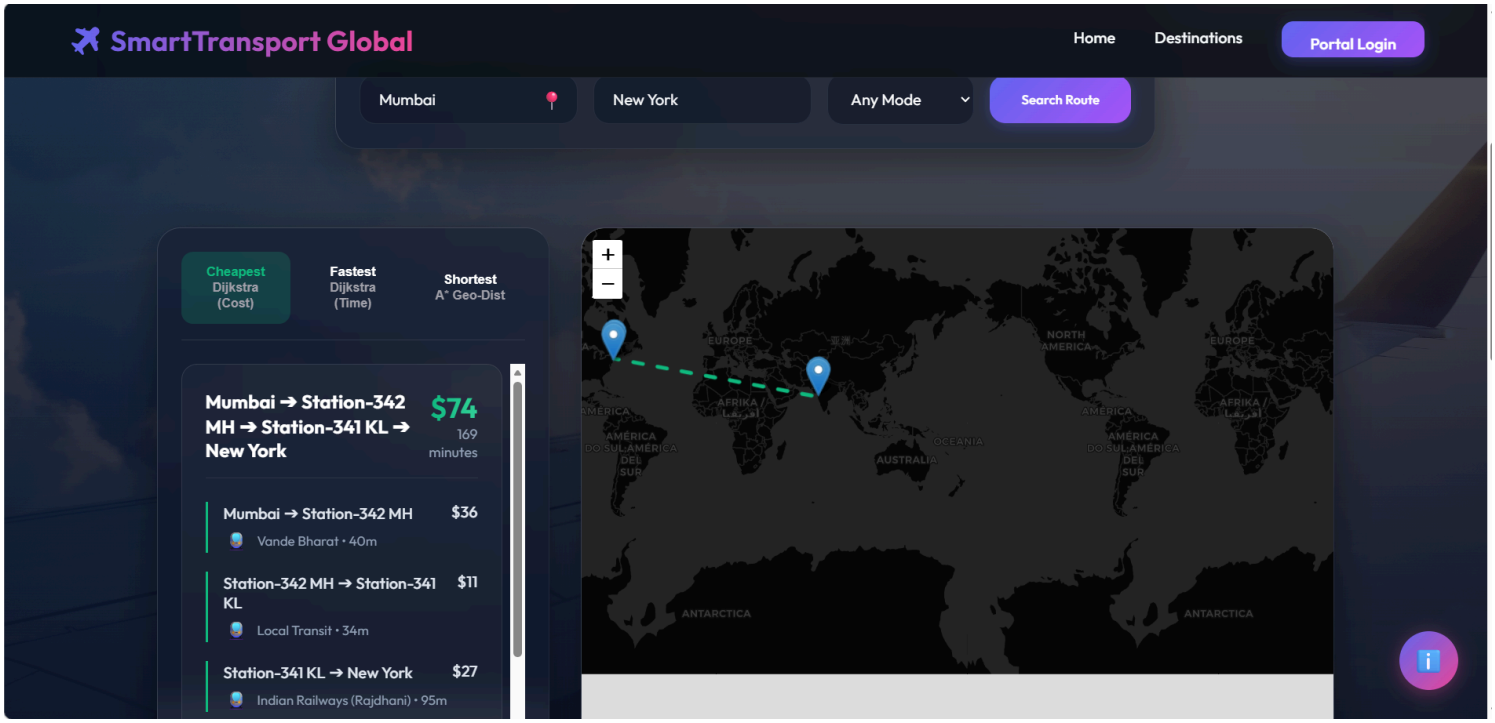
User main home page. This interface allows users to input their dynamic business details, including origin and preferred destination (e.g., INR, USD).

Figure 3



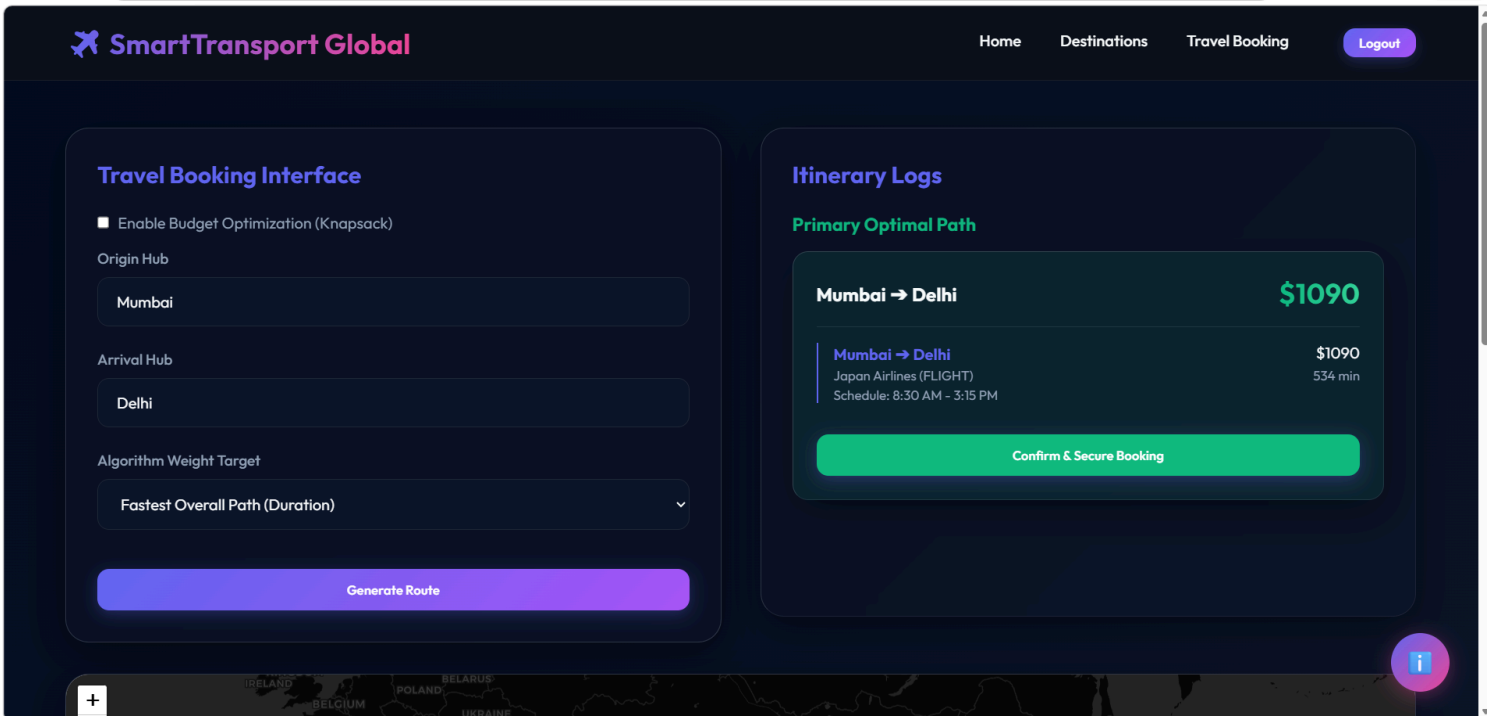
5. Output Screenshots

Figure 4



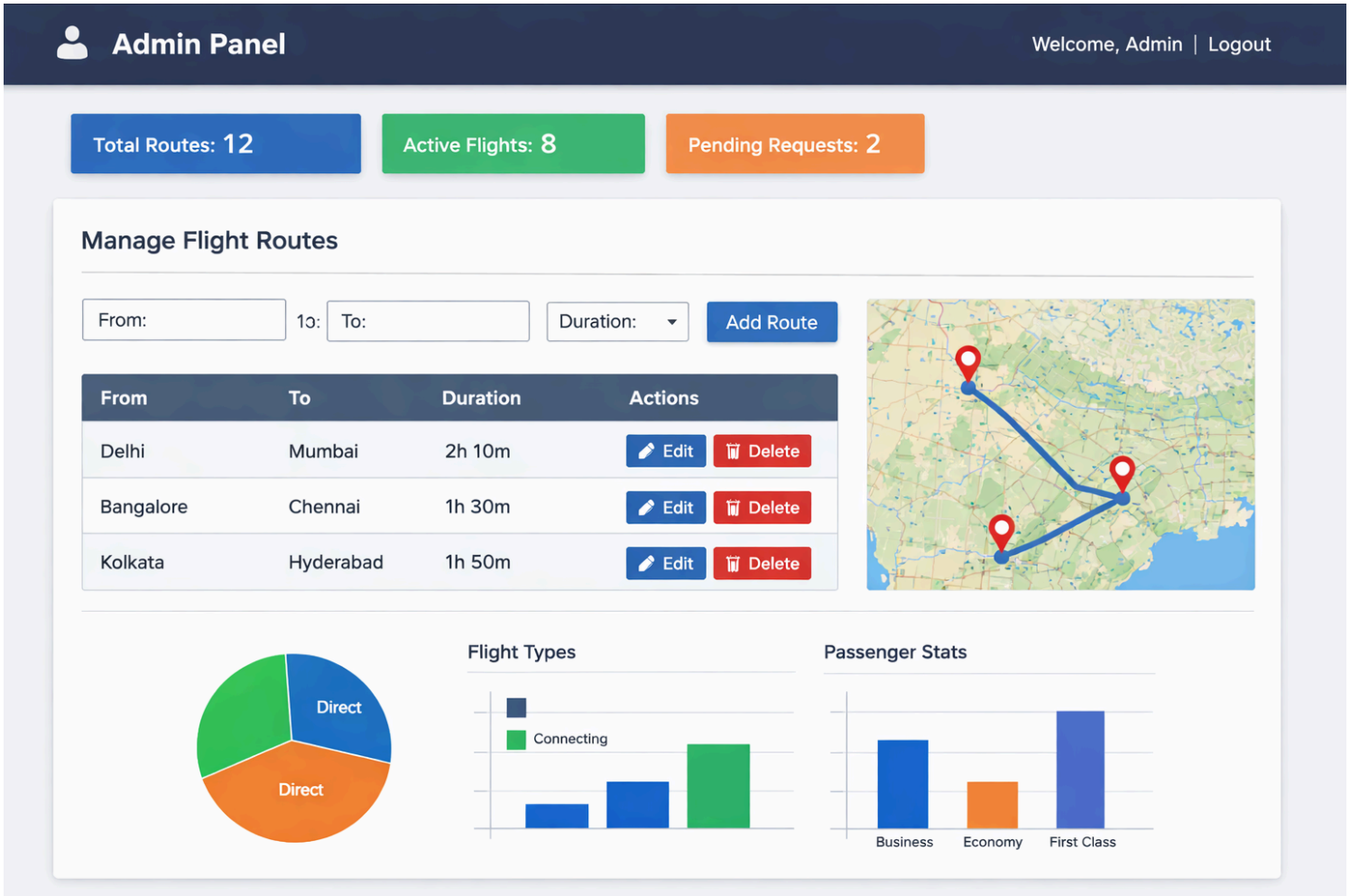
Real-time Analytics Dashboard. The output view displaying the different root he can chose and how much cost he get on it so that he can chose best root with minimum cost.

Figure 5



The system also supports advanced optimization techniques such as budget-based selection (Knapsack Algorithm), allowing users to choose routes that best fit their financial constraints while maintaining efficiency.

Figure 6



It shows the admin that how much booking and revenue generated from the app and user login and also analysis by graphical representation.

6.References

1. **React Documentation: https://react.dev**
   Used for understanding component-based architecture, hooks, and state management.
2. **React-Leaflet Documentation: https://react-leaflet.js.org**
   Used for implementing interactive maps and route visualization.
3. **Chart.js Documentation: https://www.chartjs.org**
   Used for creating dynamic charts and analytics dashboards.
4. **Node.js & Express Documentation: https://expressjs.com**
   Used for building RESTful APIs and backend services.
5. **MongoDB & Mongoose Documentation: https://mongoosejs.com**
   Used for database design and performing CRUD operations.
6. **DijkstraAlgorithmReference:**
   https://www.geeksforgeeks.org/dijkstras-shortest-path-algorithm
   Used for implementing shortest path calculation.
7. **A* Algorithm Reference: https://www.geeksforgeeks.org/a-search-algorithm**
   Used for optimized pathfinding in graph-based systems.
8. **Leaflet Maps Library: https://leafletjs.com**
   Used for rendering geographical maps and markers.
9. **MDN Web Docs: https://developer.mozilla.org**
   Used for JavaScript concepts, API handling, and web development practices.

7.Learning Outcome

Through the development of this Flight Route Optimization System, several important technical and practical concepts were learned.

● Full-Stack Development: Gained experience in building a complete web application using React for frontend and Node.js for backend.

● Algorithm Implementation: Learned how to implement graph-based algorithms such as Dijkstra and A* for solving real-world problems like route optimization.

● Data Visualization: Understood how to represent complex data using charts and maps for better user understanding.

● API Integration: Developed skills in connecting frontend with backend using RESTful APIs.

● Problem Solving: Improved logical thinking by designing efficient solutions for route calculation based on distance, cost, and time.

● UI/UX Design: Learned how to create interactive and user-friendly interfaces using modern frontend tools.

● Project Structuring: Understood how to organize a scalable project with proper architecture and modular components.